

Infrared transceiver modules for improved wireless communication

Vishay Intertechnology, Inc. has improved its TFB54xx and TFDU4xx series of infrared (IR) transceiver modules, widely used in IrDA applications for devices like energy meters, industrial controls, mobile phones, and medical



equipment. These upgraded modules now feature a PIN photodiode, infrared emitter, and a low-power control IC, boosting link distance by 20% and increasing electrostatic discharge (ESD) robustness to 2kV. They are designed for energy efficiency with low idle and shutdown mode currents and operate over broad voltage and temperature ranges.

Vishay Intertechnology, Inc.
<https://www.vishay.com>

BLE modules for Thread/Matter tech

U-blox has introduced two new Bluetooth low energy (BLE) modules, ALMA-B1 and NORA-B2, incorporating the latest nRF54 Series systems-on-chips from Nordic Semiconductor. These modules support Bluetooth LE 5.4 and the emerging Thread/Matter technologies, targeting diverse IoT applications. First one is designed for advanced applications such as industrial automation and smart homes; it features robust processing capabili-



ties suitable for indoor positioning systems. Other focuses on simple, ultra-low power applications like asset tracking.

Ublox
<https://www.u-blox.com/en>

Gate driver with augmented switching tech

Microchip Technology has introduced the 3.3kV XIFM mSiC gate driver with augmented switching technology, tai-



lored for the rising use of silicon carbide (SiC) in high-voltage applications like transportation and electric grids. This plug-and-play digital gate driver offers a compact, pre-tested solution that reduces design cycles by up to 50%, streamlining the integration process. It features 10.2kV reinforced isolation, an integrated power supply, and monitors for temperature and DC link, along with undervoltage and overvoltage lockouts, and short-circuit protection.

Microchip Technology
<https://www.microchip.com>

Dual-phase power modules for AI data centres

Infineon Technologies AG has introduced the TDM2254xD series, a line of dual-phase power modules designed to enhance the efficiency of AI data centres. As AI servers intensify global energy usage—data centers alone consume over 2% of the world's energy—the



new modules, together with XDP TM controller technology, improve voltage regulation and energy efficiency. This advancement not only decreases operating costs but also contributes to significant energy savings and reduced carbon-dioxide emissions, aligning with decarbonisation goals.

Infineon Technologies AG
<https://www.infineon.com>

Compact isolated DC/DC converter

RECOM has launched the RxxC1TF isolated DC/DC converter, designed for space-limited applications, boasting high power density and performance.



This converter is encased in a compact 5mm x 4mm 12-pad LGA package and provides regulated outputs at 3.3V or 5V with a 200mA capacity up to 80°C, extending to 125°C with derating. Targeted at sectors like industrial automation, datacomms, and IoT.

RECOM
<https://recom-power.com>

Electric double-layer supercapacitor modules

Knowles Precision Devices has introduced electric double-layer capacitor (EDLC) modules from the Cornell



Dubilier DGH and DSF Series. These modules feature a compact three-cell radial-leaded design that boosts voltage capability to 9.0 WVDC, enhancing printed circuit board efficiency. Perfect for applica-



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